AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

Chemistry Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

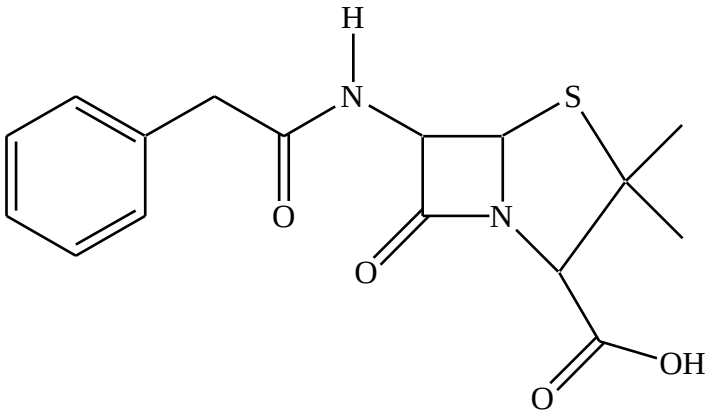
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

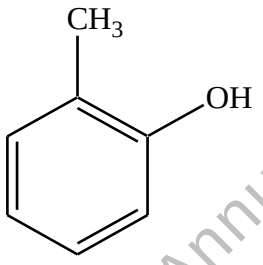
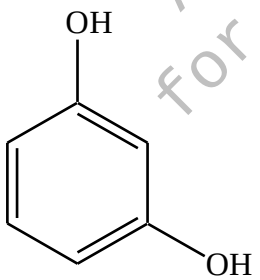
Encircle THREE chiral centres in the given structure of an organic molecule.



Q.2. (Total 4 Marks)

Complete the given table about the nomenclature of phenols.

(**Note**: IUPAC-International Union of Pure and Applied Chemistry)

S.No.	Phenol	IUPAC Name	Trivial Name
i.			
ii.			

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Q.3. (Total 4 Marks)

Write a chemical equation to show the reaction of propanal with the following reagents.

(**Note**: Mention the necessary reagents and conditions for a complete reaction.)

a. $LiAlH_4$ in dry ether (2 Marks)

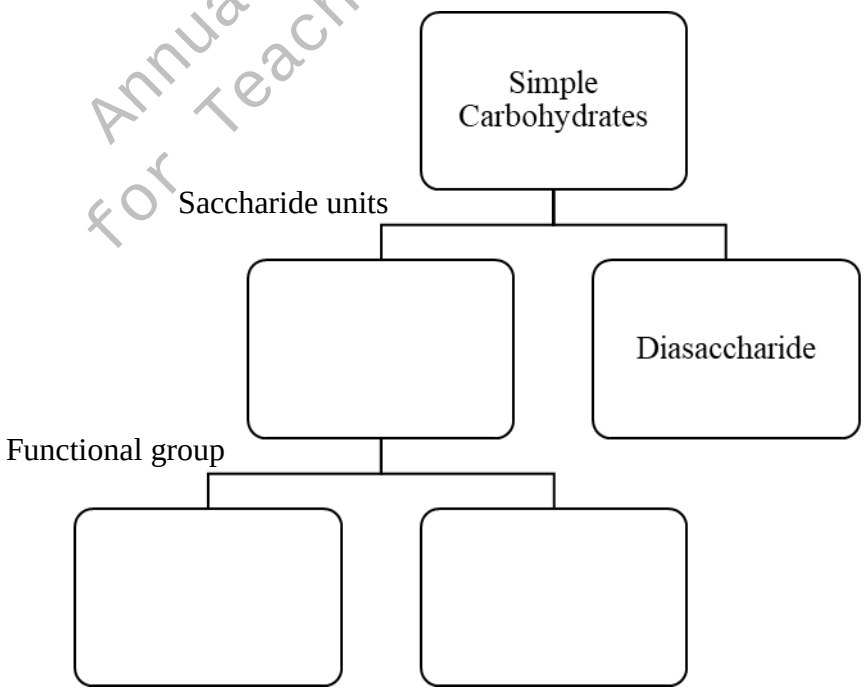
b. HCN (2 Marks)

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Q.4. (Total 3 Marks)

Complete the given classification chart based on the given attributes.

(**Note**: The first classification has been done for your convenience.)



Q.5. (Total 3 Marks)

a. Describe the process of cracking of organic compounds. (1 Mark)

b. Describe the catalytic reforming of petroleum. (2 Marks)

Q.6. (Total 4 Marks)

Write any FOUR points of significance of spectroscopic analysis in different fields of chemistry.

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Q.7. (Total 7 Marks)

EITHER

a.

i. Describe the following characteristics of alkynes. (4 Marks)

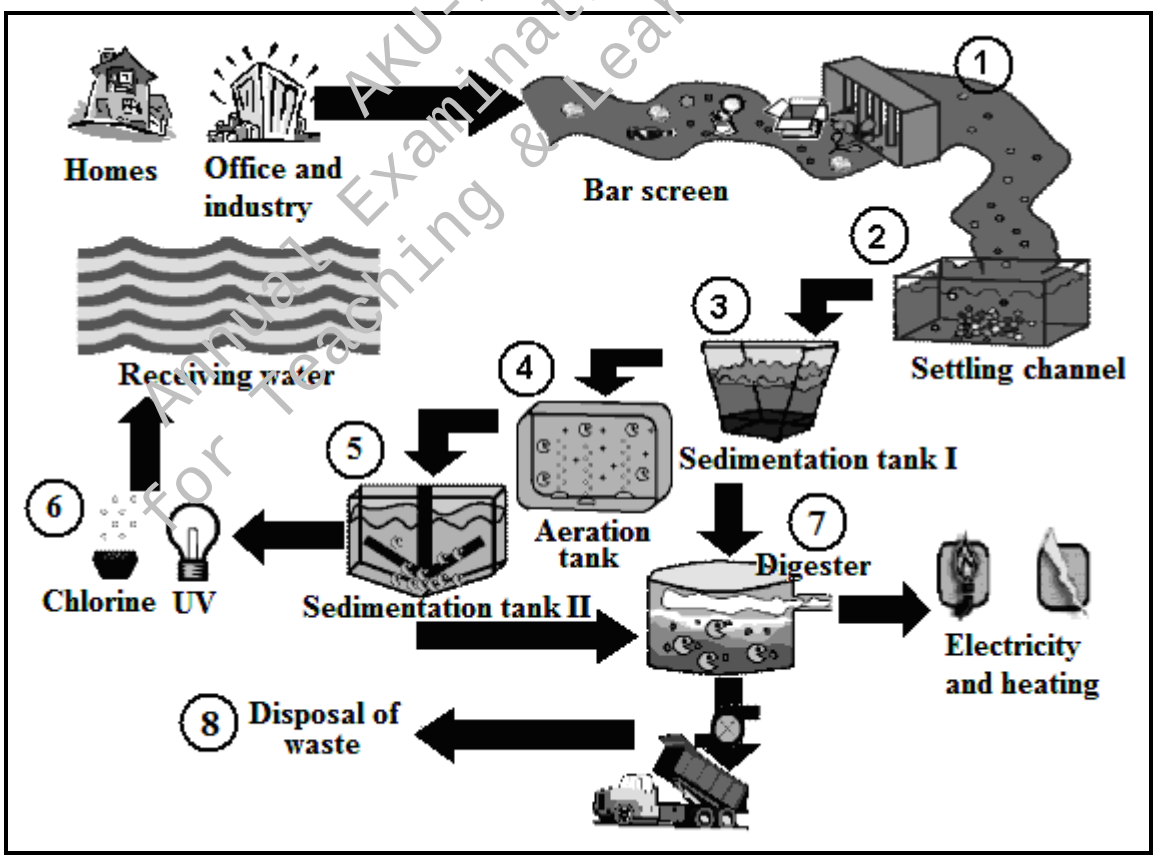
I. Reactivity towards electrophilic reagents
II. Acidic character

ii. Using balanced chemical equations, show complete reactions of ethyne (C_2H_2) with the following reactants. (3 Marks)

I. Hydrogen bromide (HBr) in accordance with Markonikov's rule
II. Chlorine gas (Cl_2) in the presence of carbon tetrachloride (CCl_4)

OR

b. The figure below shows the stages involved in sewage treatment. Using the figure below describe in detail the complete process through which sewage water is cleaned. (7 Marks)



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Q.8. (Total 7 Marks)

EITHER

a.

i. Describe the preparation of Grignard reagents in FOUR points. Support your answer with a chemical equation. (5 Marks)

ii. Give a reason why it is necessary to keep the environment inert during the synthesis of Grignard's reagent. (2 Marks)

OR

b.

i. Give a reason for each of the following statements.

I. Benzoic acid is stronger than ethanoic acid. (1 Mark)
II. The carbonyl group in a carboxylic acid is less susceptible to nucleophile than aldehyde. (2 Marks)
III. Carboxylic acids of fairly low molecular masses dissolve in water. (1 Mark)

ii. Write the general chemical equations to show the reactions of carboxylic acid with the given substances. (3 Marks)

I. Sodium metal
II. Sodium hydroxide
III. Sodium carbonate

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